Name ________________________________ **Problem Set 16**

"Make sure you do the weekly problem sets; first of all, it'll hurt your grade if you don't, and second of all it is a good way of reviewing throughout the year. It'll make the end of the year review easier." A 2019 Nerd

FR1. 1998 BC 6 **(Calculator)**

A particle moves along the curve defined by $y = x^3 - 2x$. The x-coordinate of the particle, $x(t)$, satisfies the equation $\frac{dx}{dt} = \frac{1}{\sqrt{2t+1}}$, for $t \geq 0$ with initial condition $x(0) = 3$.

a) Find $x(t)$ in terms of t.

b) Find $\frac{dy}{dt}$ in terms of t.

c) Find the position vector and speed of the particle at time $t = 4$.

FR2. 2004B BC 3 **(Calculator)**

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t	0	5	10	15	20	25	30	35	40
v(t)	7.0	9.2	9.5	7.3	4.1	2.9	2.9	4.3	7.3

A test plane flies in a straight line with positive velocity $v(t)$, in miles per minute at time t minutes, where v is a differentiable function of t. Selected values of $v(t)$ for $0 \leq t \leq 40$ are shown in the table above.

(a) Use a midpoint Riemann sum with four subintervals of equal length and values from the table to approximate $\int_0^{40} v(t)\,dt$. Show the computations that lead to your answer. Using correct units, explain the meaning of $\int_0^{40} v(t)\,dt$ in terms of the plane's flight.

(b) Based on the values in the table, what is the smallest number of instances at which the acceleration of the plane could equal zero on the open interval $0 < t < 40$? Justify your answer.

(c) The function f, defined by $f(t) = 6 + \cos\left(\frac{t}{10}\right) + 3\sin\left(\frac{7t}{40}\right)$, is used to model the velocity of the plane, in miles per minute, for $0 \leq t \leq 40$. According to this model, what is the acceleration of the plane at $t = 23$? Indicate units of measure.

(d) According to the model f, given in part (c), what is the average velocity of the plane, in miles per minute, over the time interval $0 \leq t \leq 40$?

MC1. CA For time $t \geq 0$ seconds, the position of an object traveling along a curve in the xy-plane is given by the parametric equations $x(t)$ and $y(t)$, where $\frac{dx}{dt} = t^2 + 3$ and $\frac{dy}{dt} = t^3 + t$. At what time t is the speed of the object 10 units per second?

(A) 1.675

(B) 1.813

(C) 4.217

(D) 10.191

MC2. $\int_4^\infty \frac{-2x}{\sqrt[3]{9 - x^2}}\, dx$ is

(A) $\frac{3}{2}\left(7^{\frac{2}{3}}\right)$ (B) $7^{\frac{2}{3}} + 9^{\frac{2}{3}}$ (C) $\frac{3}{2}\left(7^{\frac{2}{3}} + 9^{\frac{2}{3}}\right)$ (D) nonexistent

MC3. The region enclosed by the x-axis, the line $x = 3$, and the curve $y = \sqrt{x}$ is rotated about the x-axis. What is the volume of the solid generated?

(A) 3π (B) $2\sqrt{3}\,\pi$ (C) $\frac{9}{2}\pi$ (D) 9π

MC4. If $\frac{dy}{dx} = x^2 y$, then y could be

(A) $3\ln\left(\frac{x}{3}\right)$ (B) $e^{\frac{x^3}{3}} + 7$ (C) $2e^{\frac{x^3}{3}}$ (D) $3e^{2x}$